MPI LANCHESTER–OSIPOV

Alliances and coalitions: ally selection module

The programme's key function — assembling coalitions with allowance for reliability, force share, distance and entry delay

Document 3 of 5 · Alliances and coalitions

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Summary

Military power is relative: the outcome of a war is decided not by the strength of an individual state but by the balance of coalitions in a specific theatre. Existing ratings either ignore alliances or give “points for membership”, without distinguishing a commitment on paper from a real willingness to fight. The wars of 2022–2026 showed how different these things are: North Korea sent troops to Russia, while Russia, despite having a strategic partnership treaty with Iran, did not intervene in the 2026 war.

So in the programme alliances are not an index pillar but a **coalition builder**. The user selects a ready-made coalition (NATO, the “coalition of the willing”, the Union State, etc.) or assembles their own and sets reliability, force share, distance and entry time for each member. The programme recalculates the balance of forces and runs the battle model.

1. **Explicit model of an ally's contribution:** reliability ρ × force share s × distance decay e^(−D/λ) × entry function g(t − τ).
2. **Commitment classes**, from a collective defence treaty to an informal “axis”, with initial ρ values grounded in the practice of 2022–2026.
3. **Contribution types:** troops, supply, intelligence, territory and bases, finances. Each type affects its own part of the model.
4. **No double counting:** a coalition scenario uses the country's “autonomous power” without the “Alliances” pillar.
5. **Ready-made coalitions** for the eight assessed countries and for all potential conflict flashpoints (document 4).

1. Coalition model

1.1. Formula

```
F_coal(t) = F_own·e^(−D₀/λ₀) + Σⱼ ρⱼ · sⱼ · Fⱼ · e^(−Dⱼ/λⱼ) · g(t − τⱼ)

ρⱼ  — reliability: the probability that the ally will enter the war in this theatre (0…
1)
sⱼ  — the share of its forces allocated to the theatre (0…1)
Dⱼ  — theatre distance from its main and forward bases, thousand km;  λⱼ — projection
radius
τⱼ  — entry delay, days;  g — switch-on function (0 until τ, then a smooth rise to 1)
Fⱼ  — the ally's autonomous power (MPI without the “Alliances” pillar), converted into
conventional units
```

Converting the index into conventional units of power: M = exp(κ·MPI/100), κ = 2.6·ln 10 ≈ 6.0. This is the same span of the logarithmic scale as in the programme prototype ($N_0 = 10^{(1 + 2.6·Ñ)}$): the index is built on logarithms of count variables, so only exponentials can be added, not the scores themselves.

1.2. Contribution types

Contribution type	Where it enters the model	Example 2022–2026
Troops	Coalition N_0 (with ρ, s, D, τ)	the North Korean contingent in Russia; joint US and Israeli strikes on Iran
Supply and armaments	sustainment: reserve R and replacement rate r(t), but not N_0	supplies of weapons and equipment to Ukraine (Ramstein, purchases through allies); supplies of shells from North Korea to Russia
Intelligence and targeting	effectiveness e (multiplier)	allied satellite intelligence for Ukraine and Israel
Territory and bases	reduces D for the side's other members: D*eff = D − ρ·max(0; D − D*B), D_B — distance of the provider's territory from the theatre	US bases in the Persian Gulf; Belarusian territory in 2022; Poland and the Baltic states for NATO forces
Finances	reserve and replacement: R += η·Ffin, r += η·Ffin / τF, Ffin = 10·ρ·s·M, η = 0.15, τF = 90 days; not part of F	EU loans to Ukraine, US military aid to Israel
Air-defence coalition	defence effectiveness (interception in the salvo model)	the repulse of Iran's strike on Israel in April 2024 by the forces of several countries

The “Territory” and “Finance” types are implemented (19.10.2026, core `core/lanmil/coalition.py`, prototype). The coefficients η = 0.15 and τF = 90 days are prototype assumptions, like g(t − τ), intelligence and supply, rather than data: they can be changed, but consistently with the core. Two new scenarios show the effect: “Baltic with basing” (Poland, the Baltic states, Finland and Sweden provide territory) — on day 30 the A : B force ratio falls from 0.46 to 0.42, because Germany, the UK, France and the US deploy from bases near the theatre rather than from depth; “Russia vs. Ukraine with financing” — the ratio in F does not change, but Ukraine's reserve and replacement grow, and in the battle model the probability of a successful offensive falls noticeably while protracted fighting becomes more likely. The scale of the finance shares s is a scenario assumption.

1.3. How to avoid double counting

In the ordinary rating, the “Alliances” pillar (14 indicators) raises the MPI of members of strong alliances. If allies are added explicitly in a scenario, this pillar must be switched off, otherwise an ally will be counted twice. The programme therefore stores two values for each country: **MPI** (for the rating) and **autonomous power** (for coalitions). For the eight countries they are given in document 2.

2. Commitment classes and reliability

Class	Commitment type	Initial ρ	Examples
A	A multilateral collective defence treaty with integrated command	0.7–0.9	NATO (Art. 5)
B	Bilateral mutual assistance treaty	0.4–0.8	Russia–North Korea (2024), Pakistan–Saudi Arabia (2025), Türkiye–Azerbaijan (Shusha Declaration, 2021), US–Japan, US–Republic of Korea, US–Philippines, France–Greece (2021)
C	Strategic partnership without military commitments	0.02–0.15	Russia–Iran (2025 treaty), Russia–China, “Abraham Accords”
D	Ad hoc coalition	0.2–0.6	“Coalition of the willing” for Ukraine, Ramstein, naval coalitions in the Red Sea
E	Informal axis or network	0.1–0.5	“Axis of Resistance”, notional China–Russia–Iran–North Korea axis
F	Alliances without a military component	≈ 0	SCO, BRICS

Why ρ is not equal to 1 even for NATO. Article 5 obliges allies to take “such action as they deem necessary”, not to go to war automatically. In 2025–2026, trust within the alliance was affected by disputes over spending, tariffs and the crisis over Greenland. At the summit in Ankara (July 2026) the allies confirmed their commitments, but analysts noted limited unity. The initial ρ for the United States with respect to the European theatre is therefore 0.7, and for the European allies on the eastern flank — 0.8–0.95.

3. Catalogue of alliances and coalitions

Alliance or coalition	Members	Class	ρ	Contribution	Status as of October 2026
NATO	32 states	A	0.7–0.95	all types	Target of 5% of GDP by 2035 (The Hague, 2025); the Ankara summit (July 2026) called Russia a long-term threat
EU, Art. 42.7 TEU	27 states	B/D	0.4–0.6	finances, supply, troops	Financing of Ukraine's defence, joint procurement programmes
“Coalition of the willing”	about 30 states led by the United Kingdom and France	D	0.2 (before the truce) / 0.6 (after)	post-truce forces, supply	Paris Declaration (6 January 2026): deployment of multinational forces after a ceasefire
Ramstein (UDCG)	about 50 states	D	—	supply	Supplies and procurement of arms for Ukraine
Joint Expeditionary Force (JEF)	the United Kingdom and 9 Nordic countries	D	0.6–0.8	troops	Rapid response in Northern Europe
Union State	Russia, Belarus	A/B	0.7 (territory), 0.3 (troops)	territory, troops	Exercises near the Suwałki Gap (September 2026); nuclear weapons on the territory of Belarus
CSTO	Russia, Belarus, Kazakhstan, Kyrgyzstan, Tajikistan (Armenia has frozen its participation)	A	0.1–0.2	minimal	The organisation did not support any war of its members
Russia–North Korea treaty (2024)	Russia, North Korea	B	0.6	troops, shells, missiles	According to Seoul's estimate — about 15 thousand combat service members and more than 6 thousand engineers; an expansion is under discussion
Russia–Iran partnership (2025)	Russia, Iran	C	0.05	minimal	Russia did not intervene in the 2026 war
China–Russia	China, Russia	C	0.05 (troops)	economy, components	Economic support without military commitments
US–Israel	United States, Israel	B (de facto)	0.9	troops, supply, intelligence, missile defence	Joint campaigns against Iran in 2025 and 2026
"Axis of Resistance"	Iran, Hezbollah, the Houthis, Iraqi militias, Hamas	E	0.3–0.5	strikes, second front	Weakened; Hezbollah re-entered combat in March 2026.
US–Japan, US–Republic of Korea, US–Philippines	bilateral treaties	B	0.6–0.8	troops, bases	Tension in the South China Sea and around Taiwan
AUKUS	Australia, United Kingdom, United States	D	0.5	technology, submarines	Long-term submarine programme
Pakistan–Saudi Arabia (2025)	Pakistan, Saudi Arabia	B	0.4	troops, finances	Strategic mutual defence treaty, September 2025
Türkiye–Azerbaijan	Türkiye, Azerbaijan	B	0.6	troops, armaments	Shusha Declaration (2021)
France–Greece	France, Greece	B	0.5	troops	Strategic partnership of 2021 with a mutual assistance article
SCO, BRICS	—	F	≈ 0	—	Non-military alliances

ρ — the programme's initial values; the user changes them with a slider. The values are grounded in the behaviour of the participants in the wars of 2022–2026 and in the legal force of the commitments.

4. Ready-made coalitions for the eight countries

Country	Coalitions in the programme menu
United States	NATO · US + Israel · Indo-Pacific (Japan, Republic of Korea, Australia, the Philippines) · AUKUS · Gulf partners
Russia	Union State (Belarus) · Russia + North Korea · CSTO · Russia + Iran (partnership) · Notional China–Russia–Iran–North Korea axis
Ukraine	“Coalition of the willing” · Ramstein (supply) · EU (finances) · Poland + the Baltics + Northern Europe
Israel	US + Israel · Regional air-defence coalition (example — April 2024) · “Abraham Accords”
Iran	“Axis of Resistance” · Iran + Russia + China (no commitments)
Germany	NATO · EU (Art. 42.7) · Weimar Triangle · “Coalition of the willing”
France	NATO · EU (Art. 42.7) · France–Greece · “Coalition of the willing” · Weimar Triangle
United Kingdom	NATO · JEF · AUKUS · “Five Eyes” (intelligence) · “Coalition of the willing”

5. Worked examples

> **Status of examples.** the examples show how the algorithm works. Values for countries outside the eight (Poland, the Baltic and Nordic countries, Belarus, North Korea, the “Axis of Resistance”) are indicative and will be replaced by the full calculation. The force shares s and distances D are scenario assumptions and can be changed.

5.1. Israel and the United States against Iran (theatre — Iran)

Member	MPI without alliances	ρ	s	D, thousand km	λ	Contribution, c.u.
Israel	63.7	1.00	0.70	1.6	4.0	21.3
United States	90.6	0.90	0.20	1.0	12.0	37.6

Member	MPI without alliances	ρ	s	D, thousand km	λ	Contribution, c.u.
Iran	32.4	1.00	1.00	0.0	3.0	6.9
“Axis of Resistance” (Hezbollah, the Houthis, militias)	20.0	0.50	1.00	0.0	1.5	1.7
Russia	65.3	0.05	0.02	2.0	5.0	0.0

Result: 58.8 c.u. versus 8.6 c.u., ratio ≈ 6.8 : 1. For an air-and-missile campaign the defence coefficient is low (d = 1.3 — hasty defence and strikes), and the required ratio at α = 1.0 is 1.30 : 1 (α and d as of 13.10.2026 — from 404 CDB90 battles and validation against 10 wars, document 6, sections 7.5–7.7). Allowing for model error (σ = 2.0, section 7.8 of document 6), the probability that the advantage is sufficient is 80%. The model predicts rapid suppression of Iran's air defence and air force, which matches the course of the 2025 and 2026 campaigns. The limitations of the model here are the survivability of the underground missile arsenal and the asymmetric threat to the Strait of Hormuz. These are captured by the salvo model and survivability indicators, not by the balance in c.u..

5.2. Russia against Ukraine (theatre — Ukraine)

Member	MPI without alliances	ρ	s	D, thousand km	λ	Contribution, c.u.
Russia	65.3	1.00	0.60	0.5	5.0	27.0
North Korea (contingent in Russia)	40.0	0.60	0.10	0.5	2.0	0.5
Belarus	25.0	0.30	0.20	0.3	1.5	0.2

Member	MPI without alliances	ρ	s	D, thousand km	λ	Contribution, c.u.
Ukraine	46.8	1.00	1.00	0.0	2.5	16.5

Current situation (Ukraine's allies supply weapons but not troops): 27.8 versus 16.5 c.u., ratio ≈ 1.68 : 1. Allied supplies enter Ukraine's sustainment (reserve and replacement), not its numbers. The defence coefficient of this war is measured from its own losses (from 13.10.2026; previously it was set

by the assumption d ≈ 2.5–3 at α ≈ 1.3): in 2024–2025 Russia lost 10,084 units of equipment, Ukraine — 6,023 (Oryx), d = 1.7, and at α = 1.0 a ratio of 1.70 : 1 is required. The actual ratio falls short of the required one by hundredths, so the model predicts a positional war of attrition with slow progress by the attacker. This matches the course of the war in 2023–2026, but is not an independent validation: d is taken from the same war, and the conclusion rests on a slim margin (at d = 1.67 without rounding it changes) — document 6, section 7.7. Allowing for model error (σ = 2.0), the probability that Russia's advantage is sufficient is 50%: the model does not distinguish a positional war from a breakthrough here.

Member	MPI without alliances	ρ	s	D, thousand km	λ	Contribution, c.u.
Ukraine	46.8	1.00	1.00	0.0	2.5	16.5
United Kingdom (post-ceasefire forces)	60.2	0.60	0.10	2.0	7.0	1.7
France (post-ceasefire forces)	65.4	0.60	0.10	2.0	7.0	2.3
Germany (on adjacent territory)	56.9	0.60	0.05	1.5	4.0	0.6

After a ceasefire with the deployment of “coalition of the willing” forces: the ratio falls to ≈ 1.32 : 1. The main effect of the guarantees is not an increase in c.u. but a change in ρ: an attack on territory where British and French troops are stationed becomes an attack on nuclear powers. The programme reflects this through the deterrence axis: the “offensive” scenario objective is blocked while coalition forces with nuclear participants are in the area.

5.3. Russia and Belarus against European NATO members (theatre — the Baltics)

Member	MPI without alliances	ρ	s	D, thousand km	λ	Contribution, c.u.
Russia	65.3	1.00	0.40	0.3	5.0	18.8
Belarus	25.0	0.70	0.50	0.2	1.5	1.4

Member	MPI without alliances	ρ	s	D, thousand km	λ	Contribution, c.u.
Poland	55.0	0.95	0.60	0.3	3.0	13.9
the Baltic states, Finland, Sweden (together)	50.0	0.95	0.60	0.3	3.0	10.3
Germany	56.9	0.85	0.30	1.0	4.0	6.0
United Kingdom	60.2	0.85	0.25	1.8	7.0	6.0
France	65.4	0.80	0.25	1.8	7.0	7.8

Member	MPI without alliances	ρ	s	D, thousand km	λ	Contribution, c.u.
United States	90.6	0.70	0.15	7.0	12.0	13.3

Result: European allies against Russia and Belarus — ≈ 2.18 : 1, with US participation — ≈ 2.84 : 1. The share of Russia's forces available for this theatre is limited by the war with Ukraine (s = 0.4). However, at the outset European forces deploy with a delay τ, and Russia can act suddenly. The programme therefore

builds a balance curve over time: in the first 7–14 days the local ratio at the Suwałki Gap may favour the attacker, and by day 30 — favour NATO. This explains the priority of forward presence (Germany's brigade in Lithuania) and air defence of the eastern flank.

6. Function in the programme interface

6.1. Coalition builder

- **Selecting a ready-made coalition** from the menu (section 4) or assembling your own by dragging countries onto sides A and B.
- **For each member** — sliders for ρ (reliability), s (force share), τ (delay, days), a choice of contribution types (troops, supply, intelligence, territory, finances) and automatic calculation of D from the theatre map.
- **Hints:** the programme suggests allies under the parties' treaties and warns about a conflict of interest (for example, Greece and Türkiye in the same NATO bloc, Armenia in the CSTO).
- **“What if the ally does not join” toggle** — instant recalculation without the member.
- **Timeline** shows the entry of allies (τ) and the change in the balance by day.

6.2. Visualisation

- Balance column: the contribution of each member of a side, highlighting own forces and allies.
- Alliance graph: countries are nodes, treaties are edges (thickness = ρ, colour = commitment class).
- Theatre map with the participants' projection radii λ.
- The force ratio curve over time and the outcome probabilities from the battle model.

6.3. Data and updates

- Reference of treaties (parties, date, class, assistance article, link to the text).
- An event log of developments confirming or refuting reliability (participation in wars, exercises, deployments), with automatic correction of the initial ρ.
- Expert adjustment of ρ with mandatory justification and versioning.

Sources

- Paris Declaration of the “coalition of the willing”, 6 January 2026 (Al Jazeera, Euronews)
- NATO summit in Ankara, July 2026 (IDSA, Clifford Chance, War on the Rocks, Steptoe)
- Comprehensive Strategic Partnership Treaty between Russia and North Korea (2024); estimates of the North Korean contingent — Ministry of Defence of the Republic of Korea, HUR of Ukraine (2026)
- Comprehensive Strategic Partnership Treaty between Russia and Iran (January 2025)
- Strategic Mutual Defence Agreement between Pakistan and Saudi Arabia (September 2025)
- Euronews: Belarusian exercises near the Suwałki Gap (September 2026); agreement between the United States, Denmark and Greenland (CFR, September 2026)
- SIPRI, IISS — data on expenditure and on the participants' forces